

DECIMALES A FRACCIONES

Decimales

- Exactos Ej: 0,25 6,75 5,6666
- Periodicos
 - Puros : Ej: $5,\hat{3} = 5,3333\dots$ $6,\overline{467}$
 - Mixtos Ej: $5,04\overline{32} = 5,04323232\dots$

- DECIMALES EXACTOS A FRACCIÓN

$$0,25 = \frac{25}{100}$$

$$\begin{array}{r} 3,467 = 3467 \\ \quad \uparrow \uparrow \uparrow \\ \quad \underline{1000} \end{array}$$

$$2546,222 = \frac{2546222}{1000}$$

PERIÓDICOS Puros A FRACCIÓN

$$\overbrace{2,667} = \frac{2667-2}{999} = \frac{2665}{999}$$

$$253, \widehat{68} = \frac{25368 - 253}{99}$$

$$\rightarrow 3, \hat{4} = \frac{34-3}{9} = \frac{31}{9}$$

$0, \hat{9} = \frac{9}{9}$ ← ERROR MATEMÁTICO

• PERIÓDICOS MIXTOS A FRACCIÓN

$$\rightarrow 23,567 = \frac{23567 - 2356}{900}$$

$$\rightarrow 96,678P1 = \frac{96678P1 - 96678}{99000}$$

$$01 = 1$$

$$\rightarrow 0,86 \overbrace{4789} = \frac{864789 - 86}{999900}$$

Ejercicio 27: Transforme los números decimales en fracciones

- a) 0,7 b) 5,2 c) 80,71 d) $2,1\overline{5}$
e) $2,2\overline{5}$ f) $0,\overline{9}$ g) $0,85\overline{2}$ h) $3,6\overline{7}$
i) $78,86\overline{4}$ j) $12,1237\overline{8}$ k) $3,423\overline{25}$ l) $641,251\overline{43} = 641,2514\overline{3}$
m) 10,00001 n) $45,30\overline{1}$ o) $0,0000\overline{6}$ p) $0,00002\overline{53}$

0) $0,0000\hat{6} = \frac{6}{90000}$

$$\downarrow \quad \hat{\epsilon} \quad \hat{\epsilon} \quad \hat{\epsilon} \quad \hat{\epsilon} \quad 1 \quad -6, \hat{6} =$$

$$\underbrace{6,6}_{\uparrow} \cdot 10^{-5} = 6,6 \cdot \frac{1}{10^5} = \frac{6,6}{100000} = \frac{6,6}{100000}$$

$$6,66666 \cdot 10^{-5}$$

$$0,666666 \cdot 10^{-4}$$

$$0,0666666 \cdot 10^{-3}$$

$$0,00666666 \cdot 10^{-2}$$

$$0,000666666 \cdot 10^{-1}$$

$$\boxed{0,0000666666} \cdot 10^0 = 1$$

$$p) 0,0000253 = \frac{253 - 2}{9900000} = \frac{251}{9900000}$$

$$2,53 \cdot 10^{-5}$$

$$m) 10,00001 = \frac{1000001}{100000}$$

Ejercicio 28: Expresa cada número como fracción y opere

$$a) 0,1\bar{2} - 5,6 - 0,2\bar{3} + 3,1$$

$$b) \left(\frac{0,3}{\frac{5}{8}} - \frac{1}{\frac{0,4}{0,16}} \right)^4$$

$$0,2 = \frac{2}{10} = \frac{1}{5}$$

$$0,4 = \frac{4}{10} = \frac{2}{5}$$

$$\begin{aligned} b) \left(\frac{\frac{0,3}{5}}{\frac{0,2}{8}} - \frac{\frac{1}{0,4}}{\frac{1}{0,16}} \right)^4 &= \left(\frac{\frac{\frac{3}{10}}{5}}{\frac{\frac{1}{5}}{8}} - \frac{\frac{1}{\frac{2}{5}}}{\frac{1}{\frac{4}{25}}} \right)^4 = \left(\frac{\frac{3}{10} : 5}{\frac{1}{5} : 8} - \frac{1 : \frac{2}{5}}{1 : \frac{4}{25}} \right)^4 = \\ &= \left(\frac{\frac{3}{50}}{\frac{1}{40}} - \frac{\frac{5}{2}}{\frac{25}{4}} \right)^4 = \left(\frac{3 : \frac{1}{40}}{\frac{50}{5} : \frac{40}{4}} - \frac{\frac{5}{2} : \frac{25}{4}}{\frac{2}{1} : \frac{4}{2}} \right)^4 = \left(\frac{12}{5} - \frac{2}{5} \right)^4 = \\ &= \left(\frac{10}{5} \right)^4 = 2^4 = \boxed{16} \end{aligned}$$

$$c) \left(\frac{\left(\frac{1}{5} \right)^2 - 0,111...}{\frac{2}{3}} \right)^{-2} = \left(\frac{\frac{1}{25} - \frac{1}{9}}{\frac{2}{3}} \right)^{-2} = \left(\frac{-\frac{16}{225}}{\frac{2}{3}} \right)^{-2}$$

$$\frac{1}{25} - \frac{1}{9} = \frac{9 - 25}{225} = \frac{-16}{225}$$

$$c) \left(\frac{\frac{5}{2}}{\frac{2}{3}} \right) = \left(\frac{\frac{5}{2}}{\frac{2}{3}} \right) = \left(\frac{5}{2} \cdot \frac{3}{2} \right)$$

$$= \left(\frac{-16}{225} : \frac{2}{3} \right)^{-2} = \left(\frac{-8}{75} \right)^{-2} = \left(-\frac{75}{8} \right)^2 = \frac{5625}{64}$$

$$0,5 = \frac{5}{10} = \frac{1}{2}$$

$$0,3 = \frac{3}{10} = \frac{1}{3}$$

$$e) \left(\frac{[(0,5)^2 + 0,3333...] \cdot 1,4666...}{2,444... + 0,3333...} \right)^{-1} = \left(\frac{\left[\left(\frac{1}{2} \right)^2 + \frac{1}{3} \right] \cdot \frac{22}{15}}{\frac{22}{9} + \frac{1}{3}} \right)^{-1}$$

$$= \left(\frac{\left(\frac{1}{4} + \frac{1}{3} \right) \cdot \frac{22}{15}}{\frac{25}{9}} \right)^{-1} = \left(\frac{\frac{7}{12} \cdot \frac{22}{15}}{\frac{25}{9}} \right)^{-1} = \left(\frac{\frac{77}{90}}{\frac{25}{9}} \right)^{-1}$$

$$= \left(\frac{77}{90} : \frac{25}{9} \right)^{-1} = \left(\frac{77}{250} \right)^{-1} = \frac{250}{77}$$

$$1,46 = \frac{146}{100} = \frac{73}{50}$$

$$= \frac{132}{90} = \frac{22}{15}$$

$$\frac{22}{9} + \frac{1}{3} = \frac{22+3}{9} = \frac{25}{9}$$

Ejercicio 29: Efectúe las siguientes operaciones

a) $5\sqrt{3} - \sqrt{12}$	b) $7\sqrt{10} - \sqrt{40} + \sqrt{360}$	c) $7\sqrt{a} - 5\sqrt{x} + 12\sqrt{x} - 15\sqrt{a}$
d) $7\sqrt{28} - \sqrt{63} + 6\sqrt{7} - \sqrt[6]{7^3}$	e) $2\sqrt{75} - 4\sqrt{27} + \sqrt{48}$	f) $4\sqrt[3]{16} - 2\sqrt[3]{81} + 5\sqrt[3]{8} + \frac{1}{2}\sqrt[3]{24}$
g) $5\sqrt{\frac{1}{12}} + 2\sqrt{\frac{1}{3}} + \sqrt{\frac{1}{27}}$	h) $a + 3\sqrt{a} + 5\sqrt{a^2} + 7\sqrt{a^3}$	i) $\sqrt{\frac{25}{18}} + \sqrt{\frac{40}{27}} + \sqrt{\frac{5}{6}}$

$$a) 5\sqrt{3} - \sqrt{12} = 5\sqrt{3} - \sqrt{3 \cdot 4}$$

$$= 5\sqrt{3} - \sqrt{3} \cdot \sqrt{4} = 5\sqrt{3} - \sqrt{3} \cdot 2$$

$$= (5 - 2) \cdot \sqrt{3} = 3\sqrt{3}$$

Factorizar

60	2
30	2
15	3
5	5
1	

$60 = 2^2 \cdot 3 \cdot 5$

$$b) 7\sqrt{10} - \sqrt{40} + \sqrt{360} = 7\sqrt{10} - \sqrt{4 \cdot 10} + \sqrt{36 \cdot 10}$$

$$= 7\sqrt{10} - \sqrt{4} \cdot \sqrt{10} + \sqrt{36} \cdot \sqrt{10}$$

$$= 7\sqrt{10} - 2\sqrt{10} + 6\sqrt{10}$$

$$= (7 - 2 + 6) \sqrt{10} = 11\sqrt{10}$$

$$36 = 2^2 \cdot 3^2 = \sqrt{2^2 \cdot 3^2} \cdot \sqrt{10}$$

$$36 = 6\sqrt{10}$$

$$\sqrt{360} = \sqrt{2^2 \cdot 3^2 \cdot 2 \cdot 5} = \sqrt{2^2 \cdot 3^2} \cdot \sqrt{2 \cdot 5}$$

$$= 6\sqrt{10}$$

$$c) 7\sqrt{a} - 5\sqrt{x} + 12\sqrt{x} - 15\sqrt{a} - 2\sqrt{x} - 15\sqrt{a} = 5\sqrt{x} + 12\sqrt{x}$$

$$\begin{aligned}
 c) \quad 7\sqrt{2} - 5\sqrt{x} + 12\sqrt{x} - 15\sqrt{2} &= 7\sqrt{2} - 15\sqrt{2} - 5\sqrt{x} + 12\sqrt{x} \\
 &= (7-15)\sqrt{2} + (-5+12)\sqrt{x} \\
 &= -8\sqrt{2} + 7\sqrt{x}
 \end{aligned}$$

$$\begin{aligned}
 h) \quad 2 + 3\sqrt{2} + 5\sqrt{2^2} + 7\sqrt{2^3} &= (2+5\cdot 2) + 3\sqrt{2} + 7\cdot 2\sqrt{2} \\
 &= (1+5)2 + (3+7\cdot 2)\sqrt{2} \\
 &= 6\cdot 2 + (3+7\cdot 2)\sqrt{2}
 \end{aligned}$$

$$\sqrt{2^3} = \sqrt{2^2 \cdot 2^1} = \sqrt{2^2} \cdot \sqrt{2} = 2\sqrt{2}$$

Ejercicio 30: Demuestre que:

$$a) (14 \cdot 2^3 - 6 \cdot 2^3 + 8 \cdot 2^3)^2 = 2^{14}$$

$$b) \frac{(4^{n-1})^n}{(4^{n+1})^n} = \frac{1}{4^{2n}}$$

$$c) \sqrt[n]{\frac{4^n \cdot 6}{4^{2n+1} + 2^{4n+1}}} = \frac{1}{4}$$

$$d) \frac{81^{3/4} \cdot (3^n)^4}{9^3 \cdot 27^{n-2}} = 3^{3+n}$$

$$c) \sqrt[n]{\frac{4^n \cdot 6}{4^{2n+1} + 2^{4n+1}}} = \frac{1}{4}$$

$$\begin{aligned}
 \sqrt[n]{\frac{4^n \cdot 6}{4^{2n+1} + 2^{4n+1}}} &= \frac{\sqrt[n]{4^n \cdot 6}}{\sqrt[n]{4^{2n+1} + 2^{4n+1}}} = \frac{\sqrt[n]{4^n} \cdot \sqrt[n]{6}}{\sqrt[n]{2^{4n+2} + 2^{4n+1}}} \\
 &= \frac{4 \cdot \sqrt[n]{3 \cdot 2}}{2^4 \cdot \sqrt[n]{3} \cdot \sqrt[n]{2}} = \frac{4}{16} \cdot \frac{\sqrt[n]{3 \cdot 2}}{\sqrt[n]{3} \cdot \sqrt[n]{2}} \\
 &= \frac{4}{16} = \frac{1}{4}
 \end{aligned}$$

$$2^{4n} \cdot 2^1 = 2^{4n+1}$$

$$\sqrt[n]{2^{4n+1}} = \sqrt[n]{2^{4n}} \cdot \sqrt[n]{2^1} = 2^4 \cdot \sqrt[n]{2}$$

$$2 \cdot (2n+1) = 4n+2$$

$$\begin{aligned}
 2^{4n+2} + 2^{4n+1} &= 2^{4n+1} \cdot 2^1 + 2^{4n+1} \\
 &= (2 + 1) 2^{4n+1} \\
 &= 3 \cdot 2^{4n+1}
 \end{aligned}$$

$$\begin{aligned}
 \sqrt[n]{3 \cdot 2^{4n+1}} &= \sqrt[n]{3^1} \cdot \sqrt[n]{2^{4n}} \cdot \sqrt[n]{2^1} \\
 &= \sqrt[n]{3} \cdot \sqrt[n]{2^{4n}} \cdot \sqrt[n]{2} \\
 &= \sqrt[n]{3} \cdot 2^4 \cdot \sqrt[n]{2} \\
 &= 2^4 \cdot \sqrt[n]{3} \cdot \sqrt[n]{2}
 \end{aligned}$$



$$\sqrt[n]{2^{4 \cdot n}} = \sqrt[n]{(2^4)^n} = 2^4$$